COMPUTER NETWORKS (CN) — UNIT I

Introduction to Computer Networks & Physical Layer

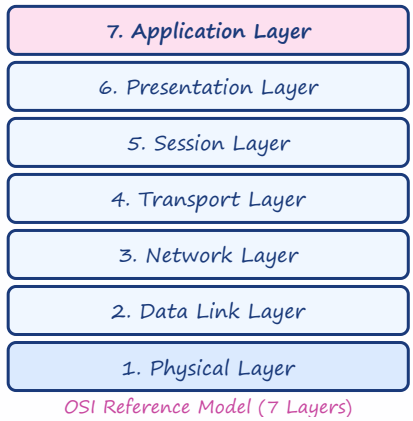
SECTION A — 2-MARK QUESTIONS & ANSWERS

1. What is the primary role of the OSI reference model in networking? *(Paper-1)*

Ans:

The **OSI (Open Systems Interconnection) Model** is a 7-layer structure that explains how data moves from one computer to another through a network.

- It divides the big job of communication into **7 small steps**. Each step (layer) has a different job.
- It allows computers from different companies to **talk to each other** using common rules.
- It helps engineers **find and fix problems** easily by checking one layer at a time.



OSI Model — 7 Layers

2. List the advantages and disadvantages of Mesh Topology? *(Paper-1)*

Ans:

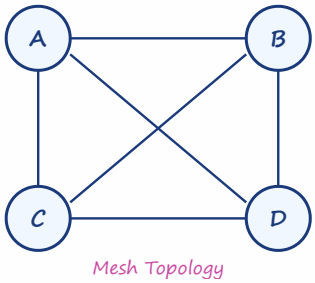
In **Mesh Topology**, every computer is connected directly to every other computer using a separate cable.

Advantages:

- Data travels on a **private wire**, so there are no traffic jams.
- If one cable breaks, the rest of the network **still works**.

Disadvantages:

- Needs **too many cables** — N computers need N(N-1)/2 cables.
- **Very expensive** and hard to install.



Mesh Topology

Mesh Topology

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3. List and briefly explain any three functions of the Network Layer in OSI. *(Paper-2)*

Ans:

The **Network Layer (Layer 3)** moves data packets from the sender to the receiver across different networks.

1. **Routing:** Finds the best path from sender to receiver.
2. **Logical Addressing:** Adds IP address of sender and receiver to each packet.
3. **Packetizing:** Breaks large data into smaller packets for sending.



Routing + IP Addressing + Packetizing
Network Layer (Layer 3)

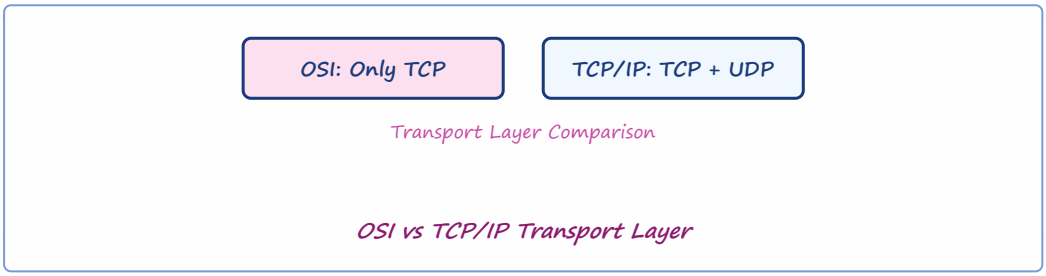
Network Layer Functions

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4. State two key differences between the OSI Transport Layer and TCP/IP Transport Layer? *(Paper-2)*

Ans:

Feature	OSI Transport Layer	TCP/IP Transport Layer
Service Type	Supports only Connection-Oriented service.	Supports both TCP (reliable) and UDP (fast).
Implementation	Only a theory — not used in real networks.	Used in real life — powers the Internet.



OSI vs TCP/IP Transport Layer

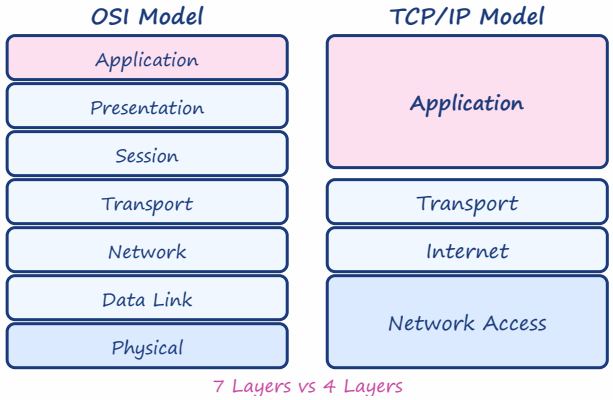
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5. Mention two key differences between OSI and TCP/IP models. *(Paper-3)*

Ans:

The **OSI Model** is a classroom guide. The **TCP/IP Model** is the real system used by the Internet.

Feature	OSI Model	TCP/IP Model
Layers	Has **7** layers.	Has **4** layers.
Approach	Theory was designed first, then protocols.	Protocols were built first, then model.



OSI (7 Layers) vs TCP/IP (4 Layers)

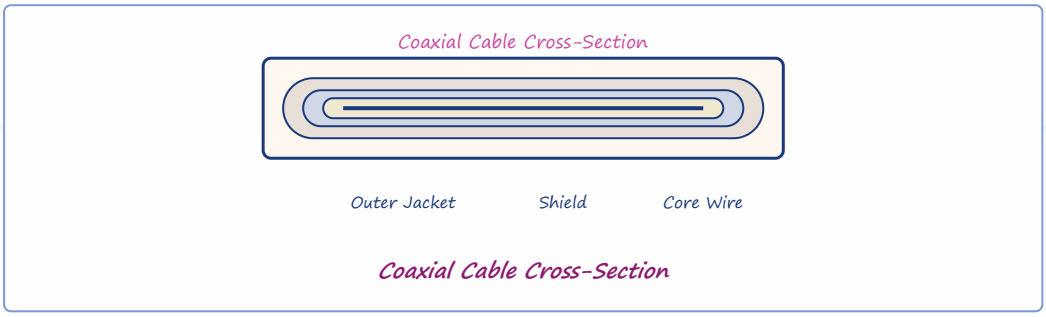
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6. What is coaxial cable and where is it commonly used? *(Paper-3)*

Ans:

A **Coaxial Cable** is a copper cable with 4 parts: center copper wire, plastic insulation, metal mesh shield, and outer cover.

- The metal shield **blocks outside noise** that can disturb the data.
- **Uses:** Cable TV connections, broadband internet modems, CCTV cameras.



Coaxial Cable Cross-Section

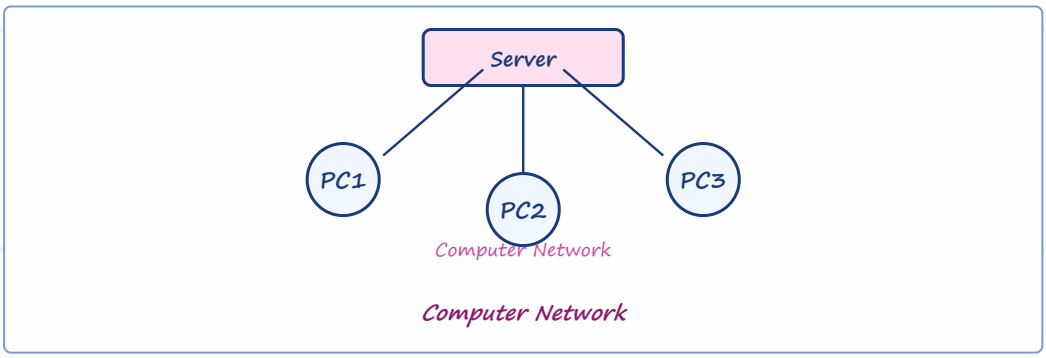
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7. Define a computer network. *(MID-1)*

Ans:

A **Computer Network** is a group of two or more computers connected by cables or wireless signals to share data, files, and devices.

- **Resource Sharing:** Many computers can share one printer or storage drive.
- **Communication:** Users can send emails and files across the world in seconds.



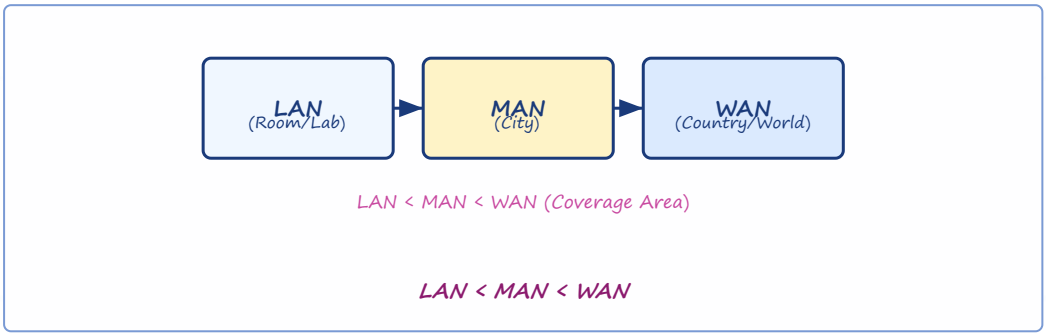
Computer Network

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8. What are LAN, MAN and WAN? *(MID-1)*

Ans:

- **LAN (Local Area Network):** Covers a small area like a room or lab (within 1 km).
- **MAN (Metropolitan Area Network):** Covers a city (5 to 50 km). Example: city cable TV.
- **WAN (Wide Area Network):** Covers a country or the whole world. Example: the Internet.



LAN < MAN < WAN

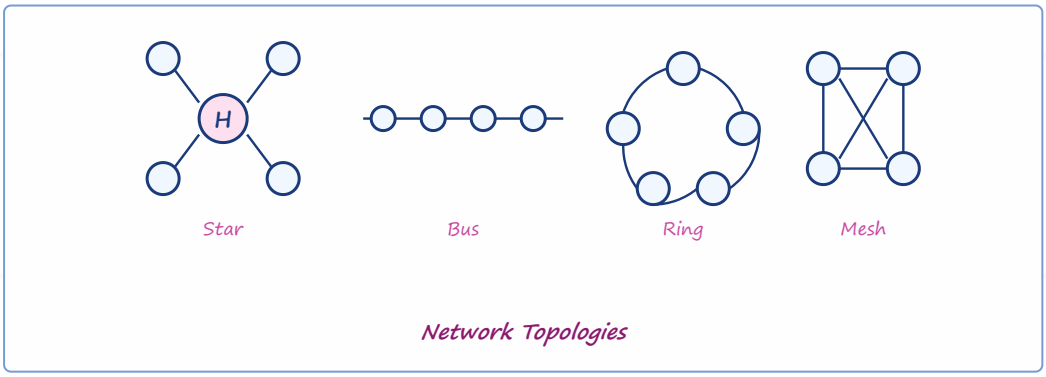
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9. List the different types of network topologies. *(MID-1)*

Ans:

A **Network Topology** is the way computers and cables are arranged.

1. **Bus:** All computers share one long cable.
2. **Star:** All computers connect to a central Hub/Switch.
3. **Ring:** Computers form a circle. Data goes in one direction.
4. **Mesh:** Every computer has a direct cable to every other.
5. **Tree:** Star networks joined together in a tree shape.



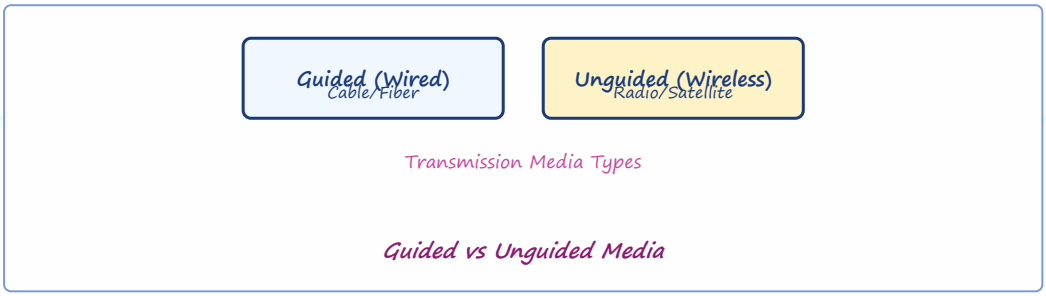
Network Topologies

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10. What is the difference between guided and unguided transmission media? *(MID-1)*

Ans:

Feature	Guided (Wired)	Unguided (Wireless)
Path	Data travels through a physical cable.	Data travels through air as waves.
Direction	Signal stays inside the wire.	Signal spreads in all directions.
Examples	Twisted pair, Coaxial, Fiber.	Wi-Fi, Bluetooth, Satellite.



Guided vs Unguided Media

SECTION B — 5-MARK QUESTIONS & ANSWERS

1. What is a Computer network? Explain the OSI model with diagram? *(Paper-1)*

Ans:

Introduction:

A **Computer Network** is a group of computers connected together by cables or wireless signals to share data and devices.

The **OSI (Open Systems Interconnection) Model** was created by ISO in 1984. It divides network communication into **7 layers**. Each layer has a specific job.

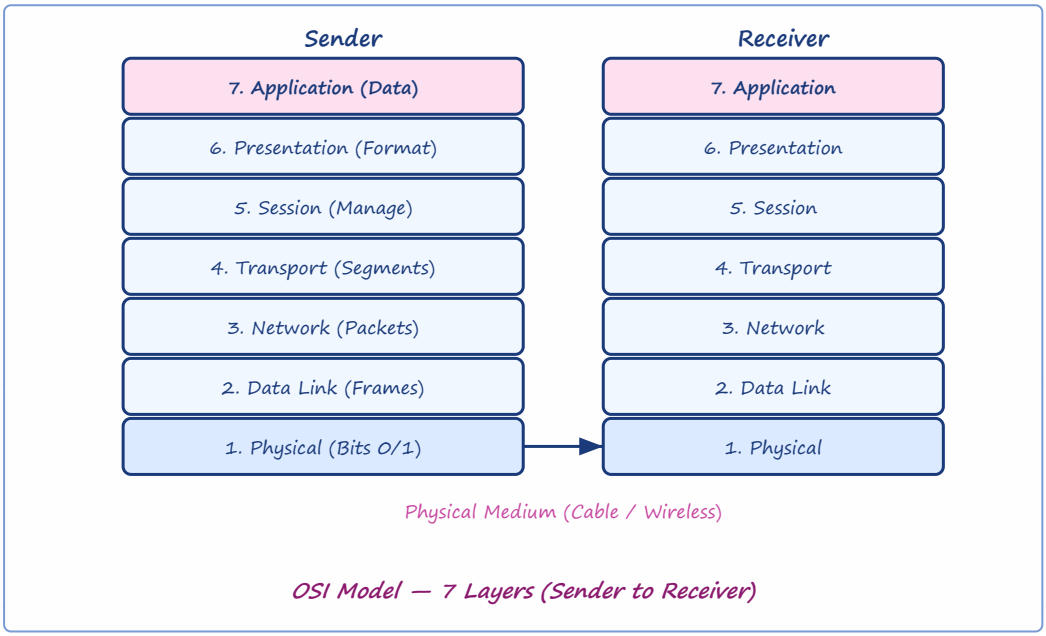
The 7 Layers of OSI Model:

1. **Application Layer (Layer 7):** This is the layer that the user directly uses. It provides services like web browsing (HTTP), email (SMTP), and file transfer (FTP).
2. **Presentation Layer (Layer 6):** It changes data into a format the receiver can understand. It also encrypts (locks) data for security and compresses (shrinks) data for faster sending.
3. **Session Layer (Layer 5):** It starts, manages, and ends the conversation between two computers. It places checkpoints so if a long download fails, it can restart from the last checkpoint.
4. **Transport Layer (Layer 4):** It delivers data to the correct application using **Port Numbers**. It breaks messages into smaller pieces called **Segments**. Protocols: TCP (reliable) and UDP

(fast).

5. **Network Layer (Layer 3):** It moves **Packets** from source to destination across networks. It adds **IP Addresses** and finds the best path using routers.
6. **Data Link Layer (Layer 2):** It puts packets into **Frames** and sends them between two directly connected devices. It adds **MAC Addresses** and checks for errors.
7. **Physical Layer (Layer 1):** It sends raw **Bits (0s and 1s)** as electrical signals, light, or radio waves through cables or air.

Diagram:



OSI Model — 7 Layers (Sender to Receiver)

Summary Table:

Layer	Data Unit	Main Job	Protocols
7. Application	Data	User services	HTTP, SMTP, FTP
6. Presentation	Data	Format, encrypt, compress	SSL/TLS, JPEG
5. Session	Data	Start/end sessions	RPC, NetBIOS
4. Transport	Segments	Process-to-process delivery	TCP, UDP
3. Network	Packets	Routing with IP address	IPv4, IPv6
2. Data Link	Frames	Hop-to-hop with MAC	Ethernet
1. Physical	Bits	Send signals on wire/air	Cables, Hubs

Conclusion:

> *The OSI model divides network communication into 7 clear steps. Each layer does one job. This makes it easy to understand, build, and fix networks.*

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2. Explain Applications of Computer networks? *(Paper-1)*

Ans:

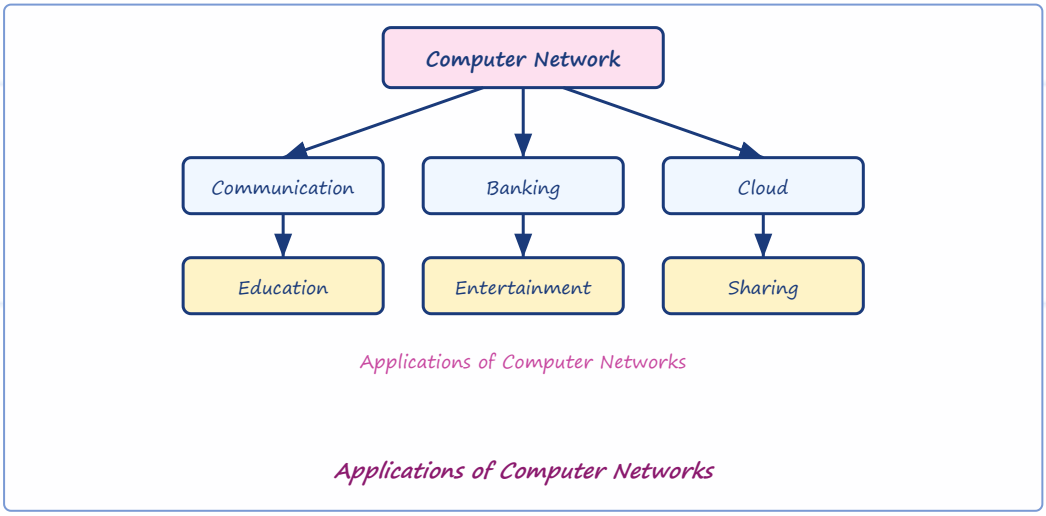
Introduction:

Computer networks connect computers across the world. They are used in almost every part of daily life.

Major Applications:

1. **Resource Sharing:** In an office, many computers can share one printer and one storage drive. This saves money.
2. **Communication:** Email (Gmail), video calls (Zoom), and messaging (WhatsApp) allow instant communication across the world.
3. **World Wide Web:** Anyone with internet can search for information using Google or read from Wikipedia.
4. **E-Commerce & Banking:** People buy products online (Amazon, Flipkart) and send money using UPI and internet banking.
5. **Cloud Computing:** Companies store their data and software on remote servers (AWS, Google Cloud). Workers can work from home using VPN.
6. **Entertainment:** People watch movies (Netflix, YouTube) and play online games using high-speed networks.

Diagram:



Applications of Computer Networks

Summary Table:

Application	Benefit	Example
Sharing	Save money by sharing devices	Office printer, shared drive
Communication	Talk to anyone instantly	WhatsApp, Zoom, Gmail
Banking	Transfer money from home	UPI, Net Banking
Education	Study from anywhere	NPTEL, Coursera
Entertainment	Watch movies, play games	Netflix, YouTube

> **Conclusion:**

> Computer networks have removed the limits of distance. People can share data, communicate, shop, study, and enjoy entertainment from anywhere in the world.

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3. List the advantages and disadvantages of Bus and Ring Topologies? *(Paper-1)*

Ans:

> **Introduction:**

A **Network Topology** is the way computers and cables are arranged. Bus and Ring are two common topologies.

> **1) Bus Topology:**

All computers connect to one long central cable called the **backbone**. Small devices called **terminators** are placed at both ends.

Advantages:

1. Needs very little cable — cheapest to install.
2. Easy to add a new computer by connecting to the main cable.

Disadvantages:

1. If the main cable breaks, the **entire network stops**.
2. Finding the exact break in a long cable is very hard.
3. When many computers send data at the same time, **collisions** happen and speed drops.

> **2) Ring Topology:**

Computers connect in a **circle**. Data moves in one direction using a special packet called a **Token**.

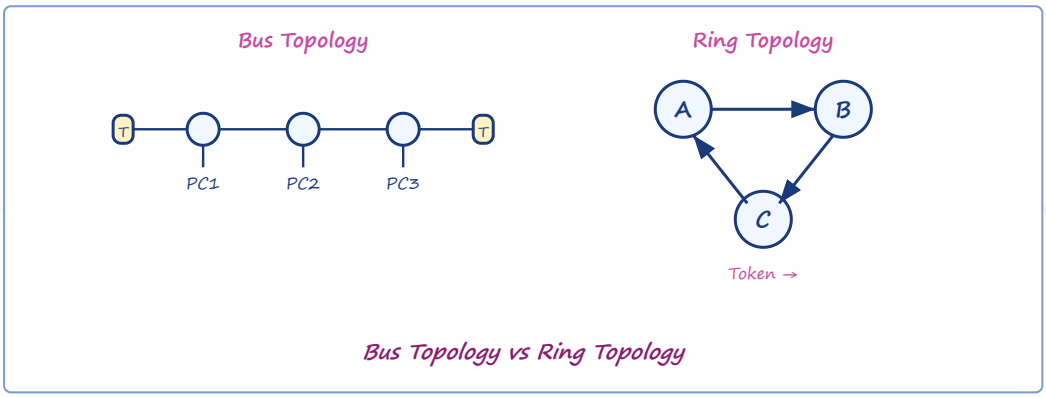
Advantages:

1. **No collisions** — only the computer with the token can send data.
2. Every computer gets a fair turn.

Disadvantages:

1. If **one computer or cable breaks**, the whole ring stops.
2. Adding or removing a computer disrupts the network.
3. Data must pass through every computer to reach the destination.

Diagram:



Bus Topology vs Ring Topology

Comparison Table:

Feature	Bus Topology	Ring Topology
Shape	Straight line	Circle
Data Flow	Both directions	One direction only
Collision	Yes, collisions happen	No, token prevents collision
Cable Break	Network splits and stops	Entire ring stops
Cost	Very low	Low to moderate

Conclusion:

> Bus topology is the cheapest but breaks easily. Ring topology is fair but also stops if one computer fails. Modern networks prefer Star topology because it is more reliable.

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4. Explain optical fiber cable with the advantages and disadvantages? *(Paper-1)*

Ans:

Introduction:

Optical Fiber Cable is the fastest communication cable. It sends data as **light pulses** through thin glass strands instead of electrical signals.

Construction:

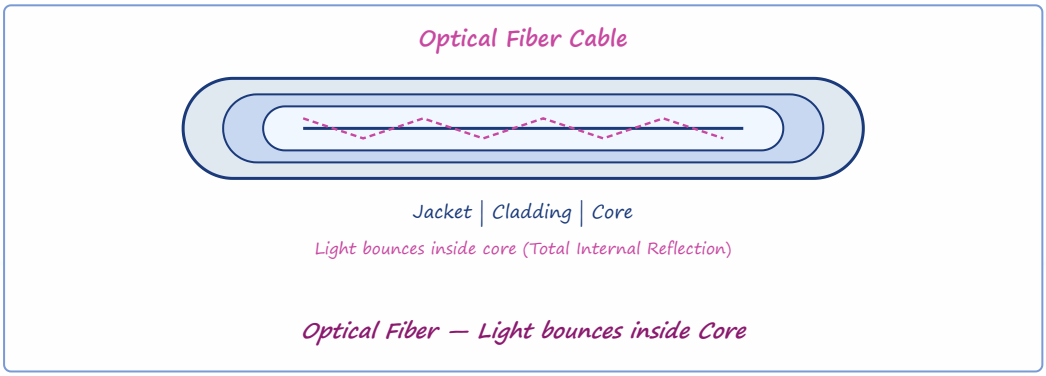
An optical fiber cable has 3 layers:

1. **Core:** The innermost thin glass where light travels.
2. **Cladding:** Glass layer around the core. It has a lower refractive index, so light stays inside the core.
3. **Jacket:** Outer plastic cover that protects the glass from damage.

Working Principle — Total Internal Reflection (TIR):

When light hits the boundary between core and cladding at a large angle, it **bounces back** into the core. This keeps happening, so light zigzags all the way to the other end at very high speed.

Diagram:



Optical Fiber — Light bounces inside Core

Types of Optical Fiber:

Type	Core Size	How Light Travels	Use
Single-Mode	8–10 microns (very thin)	Straight line	Long distance (up to 100 km)
Multi-Mode	50–62.5 microns (thicker)	Bounces at different angles	Short distance (campus, data center)

Advantages:

1. **Very high speed** — can send hundreds of Gbps of data.
2. **No electrical noise** — light is not affected by nearby power lines or motors.
3. **Long distance** — signal can travel tens of km without boosters.
4. **Very secure** — cannot be tapped without cutting the fiber.

Disadvantages:

1. **Expensive** — glass cables and laser equipment cost more.
2. **Hard to join** — needs special tools to connect two cut fibers.
3. **Fragile** — glass can break if bent too sharply.

Conclusion:

> Optical fiber is the fastest and most reliable cable. It is used for internet backbone, undersea cables, and 5G networks because of its unmatched speed and noise immunity.

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5. Explain the TCP/IP model with diagram? *(Paper-2)*

Ans:

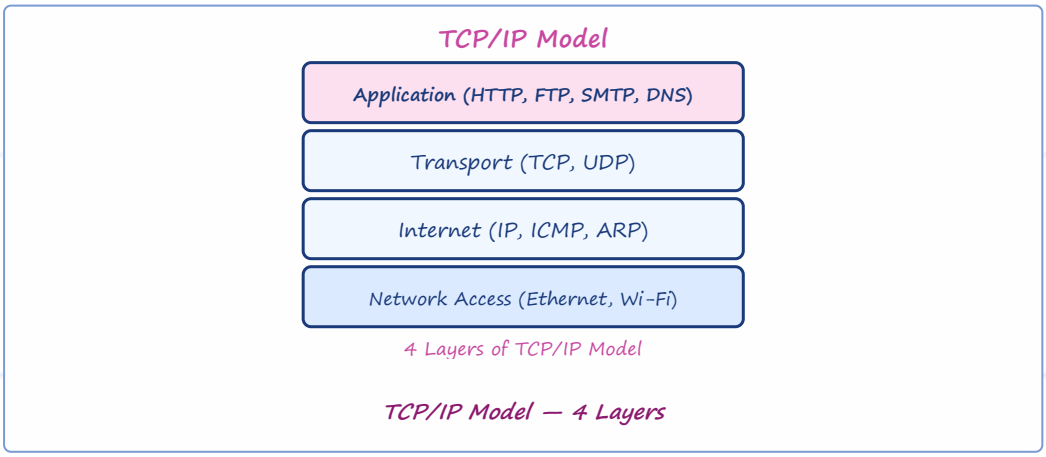
Introduction:

The **TCP/IP Model** is the actual working model of the Internet. It was created by the US Department of Defense. It has **4 layers**.

The 4 Layers:

1. **Application Layer:** This is where user programs work. It combines OSI Layers 5, 6, and 7. Protocols: HTTP (web), SMTP (email), FTP (files), DNS (domain names).
2. **Transport Layer:** It delivers data to the right program using **Port Numbers**. TCP = reliable, sends acknowledgements. UDP = fast, no acknowledgements.
3. **Internet Layer:** It adds **IP addresses** to packets and routes them from source to destination using routers. Protocols: IPv4, IPv6, ICMP.
4. **Network Access Layer:** It combines OSI Layers 1 and 2. It handles cables, MAC addresses, and puts data signals on the wire. Technologies: Ethernet, Wi-Fi.

Diagram:



TCP/IP Model — 4 Layers

Comparison — OSI vs TCP/IP:

Feature	OSI Model	TCP/IP Model
Layers	7 layers	4 layers
Approach	Theory first, then protocols	Protocols first, then model
Real Use	Used only for teaching	Powers the real Internet
Transport	Only connection-oriented	TCP + UDP

Conclusion:

> The TCP/IP model is the backbone of the Internet. Every smartphone, computer, and server uses TCP/IP to communicate.

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6. Explain about coaxial cable with the advantages and disadvantages? *(Paper-2)*

Ans:

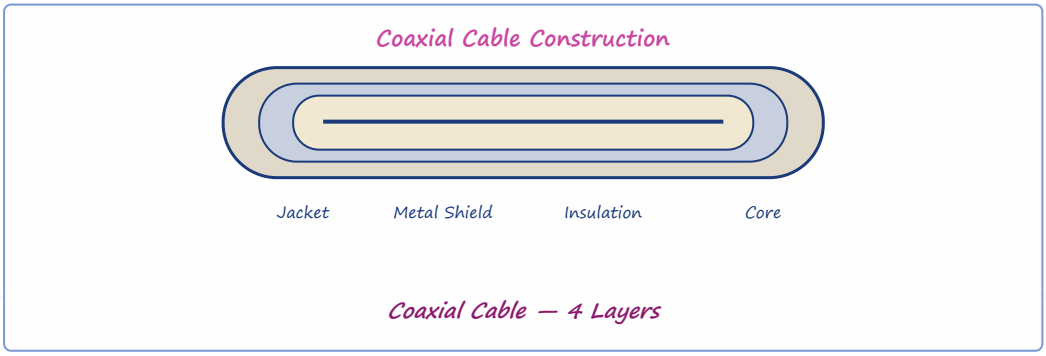
Introduction:

A **Coaxial Cable** (coax) is a round copper cable used to carry high-speed signals. It is commonly used for cable TV and broadband internet.

Construction (4 Parts):

1. **Central Copper Wire:** Carries the data signal.
2. **Plastic Insulation:** Keeps the copper wire in the center.
3. **Metal Mesh Shield:** Blocks outside electrical noise.
4. **Outer Plastic Jacket:** Protects the cable from physical damage.

Diagram:



Types:

Cable	Use
RG-59	Short distance cable TV, CCTV

RG-6	Digital satellite TV, broadband
RG-58	Old Ethernet networks (10BASE2)

Advantages:

1. **Good shielding** — metal mesh blocks outside noise better than twisted pair.
2. **Longer range** — can carry signals farther without boosters.
3. **High speed** — supports many TV channels at once.

Disadvantages:

1. **Heavy and stiff** — hard to pull through narrow walls.
2. **More expensive** than twisted pair cable.
3. **Bending breaks the shield** — noise leaks in if cable is crushed.

Conclusion:

> Coaxial cable is good for cable TV and broadband. It is stronger than twisted pair but has been replaced by Ethernet and fiber in most computer networks.

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7. What is Network topology? Explain different types of network topologies? *(Paper-2)*

Ans:

Introduction:

A **Network Topology** is the physical arrangement of computers, cables, and devices in a network.

Types of Network Topologies:

1. Star Topology (Most Popular):

- All computers connect to one central device (Hub or Switch).
- If one computer fails, only that computer stops. Others keep working.
- If the central switch fails, the entire network stops.

2. Bus Topology:

- All computers share one long backbone cable with terminators at both ends.
- Cheapest topology — uses the least cable.
- If the cable breaks, the whole network fails.

3. Ring Topology:

- Computers connect in a circle. Data moves in one direction using a token.

- No data collisions. Every computer gets a fair turn.
- If one computer or cable breaks, the ring is broken.

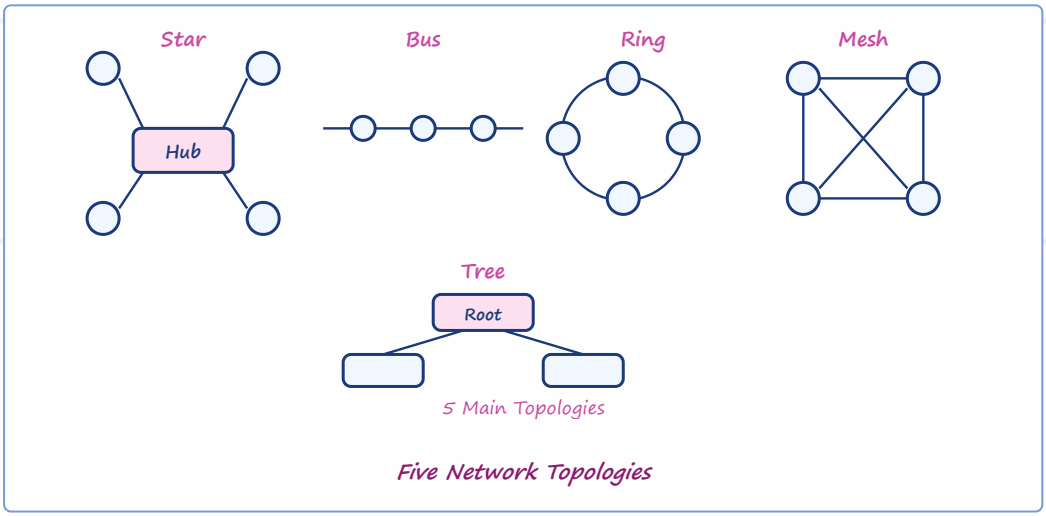
4. Mesh Topology:

- Every computer has a direct cable to every other computer.
- Most reliable — if one cable breaks, other paths still work.
- Very expensive — needs N(N-1)/2 cables for N computers.

5. Tree Topology:

- Groups of star networks connected together in a tree shape.
- Easy to expand across floors in a building.
- If the root switch fails, branches disconnect.

Diagram:



Five Network Topologies

Comparison Table:

Topology	Cable Need	Cost	Reliability	Cable Break Effect
Star	Moderate	Low-Medium	High	Only one PC stops
Bus	Very Low	Lowest	Low	Whole network stops
Ring	Low	Low	Low	Whole network stops
Mesh	Very High	Highest	Maximum	No effect
Tree	High	Medium-High	Moderate	One branch stops

Conclusion:

> *Star topology is the most popular for offices and homes because it is reliable and affordable. Mesh is used where reliability is critical, like in military or hospital networks.*

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8. List the differences between LAN, MAN, WAN? *(Paper-2)*

Ans:

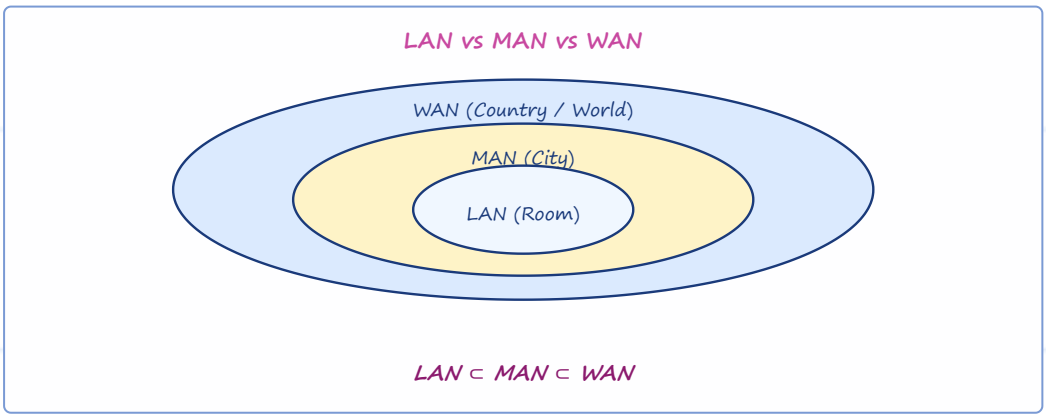
Introduction:

Networks are classified by the area they cover. The three main types are **LAN**, **MAN**, and **WAN**.

Difference Table:

Feature	LAN	MAN	WAN
Full Form	Local Area Network	Metropolitan Area Network	Wide Area Network
Area	Room, lab, building (≤ 1 km)	City or town (5–50 km)	Country or world
Speed	Very fast (100 Mbps to 10 Gbps)	Moderate (10–100 Mbps)	Slower due to distance
Owner	Private (one person or company)	Telecom company or city	Multiple providers
Delay	Very low	Moderate	High
Cost	Cheap	Higher	Very expensive
Technology	Ethernet, Wi-Fi	Fiber ring, Cable TV	Satellite, undersea fiber
Example	College lab network	City cable TV	The Internet

Diagram:



Conclusion:

> LAN is for small areas (rooms), MAN is for cities, and WAN connects the whole world. The Internet is the biggest WAN.

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9. Explain the following network topologies with diagrams, advantages, and disadvantages: Ring topology and Mesh topology. *(Paper-3)*

Ans:

Introduction:

Ring Topology connects computers in a circle. **Mesh Topology** connects every computer directly to every other computer.

1) Ring Topology:

Every computer connects to exactly two neighbors, forming a closed loop. A small packet called a **Token** moves around the ring. Only the computer holding the token can send data.

Advantages:

1. No data collisions — token controls who can send.
2. Every computer gets an equal turn.
3. Works well even with heavy traffic.

Disadvantages:

1. If one computer or cable breaks, the whole ring stops.
2. Adding a new computer needs cutting the cable temporarily.
3. Data must pass through every computer in between.

2) Mesh Topology:

Every computer has a **dedicated cable** to every other computer. For N computers, the number of cables = $N(N-1)/2$.

Example: For 6 computers: Cables = $6\times5/2$ = **15 cables**. Each computer needs 5 network ports.

Advantages:

1. Most reliable — if one cable breaks, other paths still work.
2. No traffic jams — each pair has a private wire.
3. Very secure — no one else can see your data.

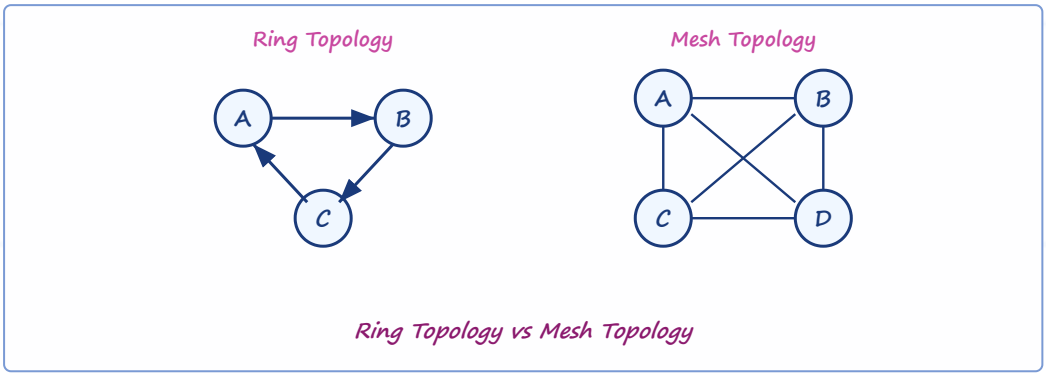
Disadvantages:

1. Very expensive — needs many cables and ports.

2. Hard to install — too many wires to manage.

3. Wasted ports — each computer needs many network cards.

Diagram:



Ring Topology vs Mesh Topology

Comparison Table:

Feature	Ring	Mesh
Cables (N PCs)	N cables	N(N-1)/2 cables
Failure Effect	Whole network stops	Only that one link fails
Collision	None (token)	None (private cables)
Cost	Low	Very high
Use	Token Ring LANs	Military, hospitals

Conclusion:

> Ring topology is simple and fair but fragile. Mesh topology is the most reliable and secure but very costly. Mesh is used where networks must never fail.

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10. Explain the TCP/IP Reference Model and compare it with the OSI Reference Model.

(MID-1)

Ans:

Introduction:

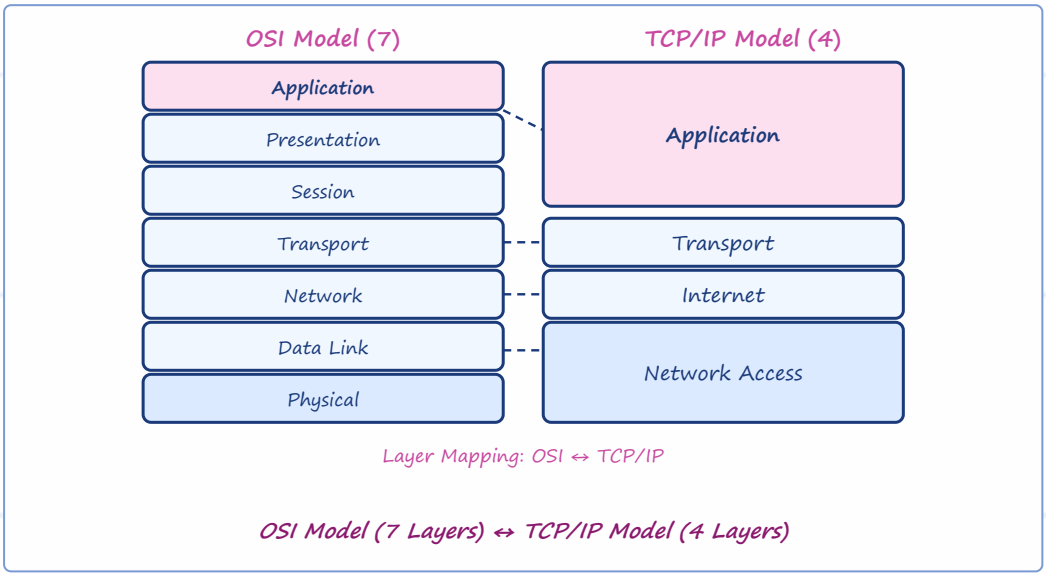
The **OSI Model** has 7 layers and was made for teaching. The **TCP/IP Model** has 4 layers and is the real system that runs the Internet.

TCP/IP Layers Explained:

1. **Application Layer:** Combines OSI layers 5, 6, 7. Provides services like HTTP, SMTP, DNS to user programs.

2. **Transport Layer:** Delivers data to the correct program. TCP = reliable. UDP = fast.
3. **Internet Layer:** Routes packets from source to destination using IP addresses.
4. **Network Access Layer:** Combines OSI layers 1 and 2. Handles cables, MAC addresses, and physical signals.

Diagram:



Comparison Table:

Feature	OSI Model	TCP/IP Model
Layers	7	4
Created by	ISO (for teaching)	DARPA (for military use)
Approach	Theory first, protocols later	Protocols first, model later
Transport	Only connection-oriented	TCP (reliable) + UDP (fast)
Network	Connection + connectionless	Only connectionless (IP)
Real Use	Not used in real networks	Powers the entire Internet

Conclusion:

> OSI is great for understanding network theory. TCP/IP is the actual model that every device in the world uses to communicate.

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11. Explain guided transmission media with reference to twisted-pair cable, coaxial cable and fiber-optic cable. *(MID-1)*

Ans:

Introduction:

Guided Media (also called **Wired Media**) uses physical cables to send data. The signal travels inside the wire. There are 3 main types.

1) Twisted-Pair Cable:

- **Construction:** Two insulated copper wires twisted around each other.
- **Why twisted?** Twisting reduces outside noise. Both wires pick up equal noise, and the receiver cancels it out.
- **Types:**
  - **UTP (Unshielded):** Cheap, used in home and office Ethernet (Cat5e, Cat6).
  - **STP (Shielded):** Has a metal shield, used in noisy factories.

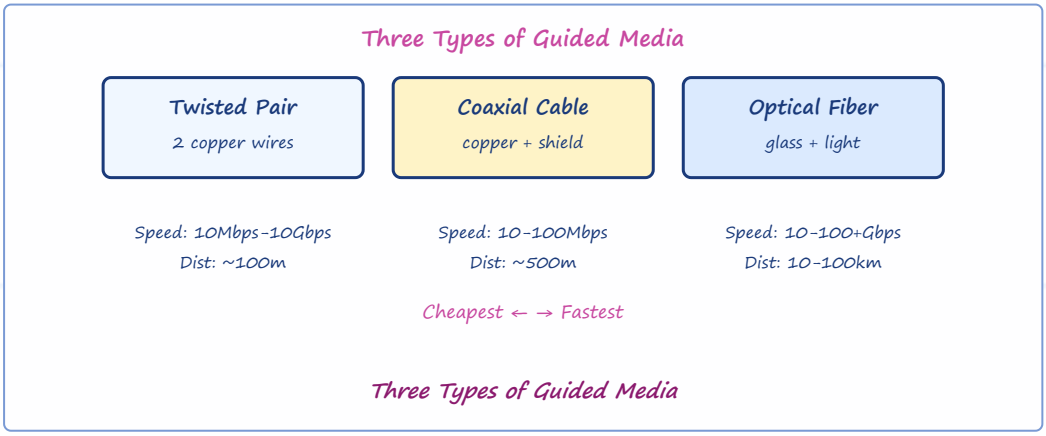
2) Coaxial Cable:

- **Construction:** Central copper wire + plastic insulation + metal mesh shield + outer jacket.
- **Working:** Central wire carries the signal. Metal shield blocks outside noise.
- **Uses:** Cable TV, CCTV cameras, broadband modems.

3) Fiber-Optic Cable:

- **Construction:** Thin glass core + glass cladding + protective jacket.
- **Working:** Data travels as **light pulses** inside the glass core. Light bounces back and forth using **Total Internal Reflection**.
- **Uses:** Internet backbone, undersea cables, 5G networks.

Diagram:



Three Types of Guided Media

Comparison Table:

Feature	Twisted Pair	Coaxial	Fiber Optic
Signal	Electrical	Electrical	Light pulses

Material	Copper wire	Copper wire	Glass/Plastic
Speed	10 Mbps – 10 Gbps	10 – 100 Mbps	10 – 100+ Gbps
Distance	~100 m	~500 m	10 – 100 km
Noise Protection	Moderate	High	Immune (zero noise)
Cost	Cheapest	Moderate	Most expensive

Conclusion:

> Twisted pair is best for offices (cheap). Coaxial is best for cable TV (good shielding). Fiber optic is best for long-distance, high-speed communication (fastest and most secure).